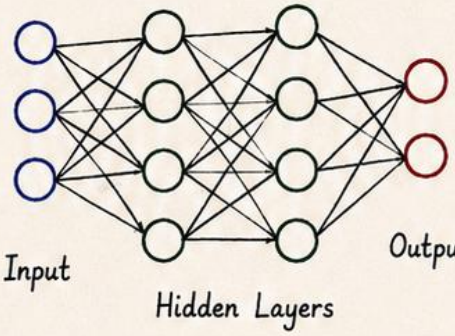
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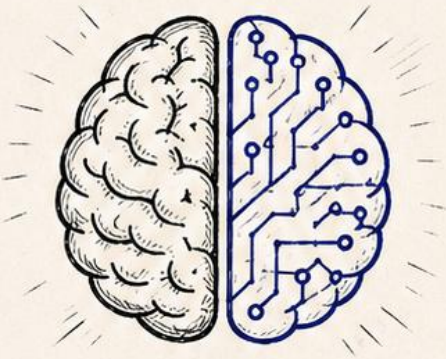
DEEP LEARNING HANDBOOK v1.0 ★

DEEP LEARNING

– HANDWRITTEN HANDBOOK –

Master the Concepts. Build Intelligent Systems.
Shape the Future with Deep Learning. ☺





"Learn Deep.
Think Big.
Build Smart.
Change the World."

Feature Learning • Representation Learning • Automation • Scalability • Innovation

What's Inside ?

1. Introduction to Deep Learning
2. Neural Networks Basics
3. Activation Functions
4. Loss Functions
5. Optimization Algorithms
6. Convolutional Neural Networks (CNN)
7. Recurrent Neural Networks (RNN)
8. Long Short-Term Memory (LSTM)
9. Transformers & Attention
10. Generative Models (GAN, VAE)
11. Deployment & Best Practices

+ Cheat Sheet & Interview Q&A

Deep Learning by the Numbers (Approx)

Active Developers	2.5+ Million
Popular Frameworks	TensorFlow, PyTorch, Keras, JAX
Model Training Data	GBs to TBs
Training Time	Minutes to Days
GPU Memory	4GB to 80GB+
Accuracy Achievable	90% to 99%+

Topics Covered

- ✓ Neural Networks
- ✓ CNN & Computer Vision
- ✓ RNN, LSTM & GRU
- ✓ Transformers & NLP
- ✓ Activation Functions
- ✓ Loss Functions
- ✓ Optimization
- ✓ Regularization
- ✓ Transfer Learning
- ✓ Generative Models
- ✓ Model Evaluation
- ✓ Deployment & MLOps
- ✓ Best Practices

Who is this for?

- ★ Students & Beginners
- ★ ML Engineers & Developers
- ★ Anyone passionate about AI

"The best way to predict the future is to invent it with Deep Learning.

DEEP LEARNING

The Next Step After ML Basics in the AI Lane

> "Deep Learning is a subset of Machine Learning inspired by the structure and function of the human brain called Artificial Neural Networks."

★ 1. WHAT IS DEEP LEARNING ?

- Deep Learning (DL) is a part of Machine Learning (ML) that uses Artificial Neural Networks (ANNs) with many layers (hidden layers) to learn patterns from data.
- It can automatically learn features from raw data without manual feature engineering.
- It works exceptionally well on large amounts of data and complex problems.

★ 2. WHY DEEP LEARNING ?

- Handles unstructured data like images, audio, text, video.
- Automatically learns hierarchical features.
- High accuracy in complex tasks.
- Scales well with big data and computational power.
- Power behind modern AI applications.

3. ML vs DL

Aspect	Machine Learning (ML)	Deep Learning (DL)
Feature Engineering	Requires manual feature engineering	Automatically learns features from data
Data Requirement	Works on small to medium datasets	Needs large amount of data for best performance
Model	Shallow models (e.g. Linear Regression, SVM, Decision Tree)	Deep Neural Networks with many hidden layers
Performance	Good for simple problems	Excellent for complex problems
Computation	Less computationally intensive	High computational power (GPU/TPU)
Examples	Spam detection, House price prediction	Image recognition, Speech recognition, Self-driving cars, ChatGPT, etc.

DL in Action

- Image Recognition
- Speech Recognition
- Self-Driving Cars
- Chatbots & NLP

★ 4. REAL LIFE APPLICATIONS

- YouTube / Netflix Recommendations
- E-commerce Recommendations
- Voice Assistants (Siri, Alexa)
- Spam Email Detection
- Medical Image Analysis
- AI Chatbots (ChatGPT)

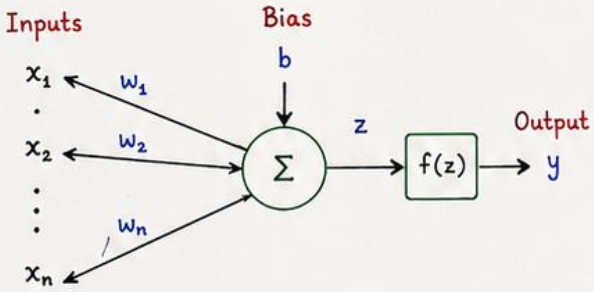
MISSION: Evergreen DSA notes - how to think in code under constraints.

2. NEURAL NETWORK FUNDAMENTALS

Deep Learning models are built using Artificial Neural Networks (ANNs) which are inspired by the human brain.

★ 2.1 NEURON (PERCEPTRON)

A neuron is the basic building block of a neural network.



x_i = inputs
w_i = weights
b = bias
z = weighted sum

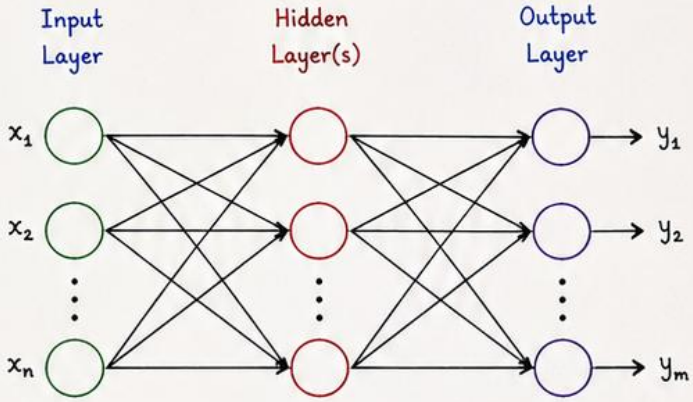
f(z) = activation function

Mathematically,

$$z = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b$$
$$y = f(z)$$

★ 2.2. LAYERS IN A NEURAL NETWORK

- Input Layer : Receives raw data.
- Hidden Layer(s) : Performs computation and extracts features.
- Output Layer : Produces the final prediction.



A network can have multiple hidden layers.
More layers → learn more complex patterns.

★ 2.3 WEIGHTS & BIASES

- Weights (w_i) control the strength of connections between neurons.
- Bias (b) allows the model to shift the activation function.
- During training, weights and biases are learned to minimize error.

★ 2.4 TYPES OF NEURAL NETWORKS (EXAMPLES)

Feedforward NN	Information moves only forward (Input → Output).
CNN (Convolutional NN)	Used for image data.
RNN (Recurrent NN)	Used for sequential / time series data.
LSTM / GRU	Special types of RNN that handle long-term dependencies.
Transformers	Used in NLP, Computer Vision, and many modern AI models.

★ 2.5 HOW A NEURAL NETWORK MAKES A PREDICTION ?

1. **Input** — Raw data is fed into the network.
2. **Forward Propagation** — Data moves through layers, each neuron performs (w.x + b) and activation.
3. **Output** — Network produces prediction.
4. **Error Calculation** — Compare prediction with actual value using loss function.
5. **Backpropagation** — Error is sent backward and weights & biases are updated.

This process repeats for many iterations until the model learns!

Tip: Think of a neural network as a function approximator that learns the mapping between Input (X) and Output (Y) from data.

Next Page → Activation Functions

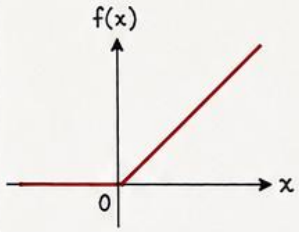
3. ACTIVATION FUNCTIONS

Activation function adds non-linearity to the neural network, enabling it to learn complex patterns.

★ Without activation function, a neural network is just a linear regression model.

① ReLU
(Rectified Linear Unit)

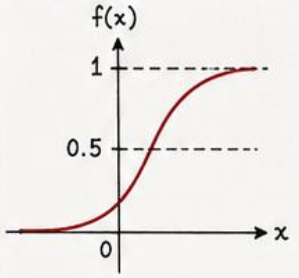
$$f(x) = \max(0, x)$$



- Range: $[0, \infty)$
- Pros:
  - Simple & fast
  - Helps reduce vanishing gradient
- Cons:
  - Dying ReLU problem (neurons can die)
  - Not zero-centered
- Used in: Hidden layers

② Sigmoid

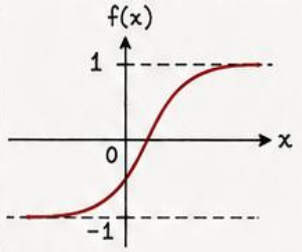
$$f(x) = \frac{1}{1 + e^{-x}}$$



- Range: $(0, 1)$
- Pros:
  - Output between 0 and 1
  - Useful for probabilities
- Cons:
  - Vanishing gradient
  - Not zero-centered
- Used in:
  - Output layer (binary classification)
  - Gates in LSTM/GRU

③ Tanh
(Hyperbolic Tangent)

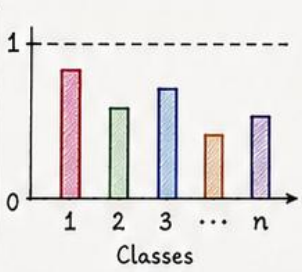
$$f(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$



- Range: $(-1, 1)$
- Pros:
  - Zero-centered
  - Smoother than sigmoid
- Cons:
  - Vanishing gradient
- Used in:
  - Hidden layers
  - Older networks (now mostly ReLU)

④ Softmax
(Multi-class output)

$$f(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}$$



- Range: $(0, 1)$ and $\sum_{i=1}^{n} f(z_i) = 1$
- Pros:
  - Converts scores to probabilities
  - Used for multi-class classification
- Cons:
  - Not used in hidden layers
- Used in:
  - Output layer for multi-class problems

★ COMPARISON SUMMARY

Function	Formula	Range	Zero-Centered?	Best Used In
ReLU	$\max(0, x)$	$[0, \infty)$	No	Hidden Layers (Default)
Sigmoid	$\frac{1}{1 + e^{-x}}$	$(0, 1)$	No	Output (Binary), Gates
Tanh	$\frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$	$(-1, 1)$	Yes	Hidden Layers
Softmax	$\frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}$	$(0, 1)$ (sum = 1)	No	Output (Multi-class)

TIPS

- Use ReLU in hidden layers for better training.
- Use Softmax for multi-class output.
- Choose activation functions based on your problem type!

☆ Remember: The right activation function can make a huge difference in how well your model learns.

4. FORWARD & BACKPROPAGATION

Forward Propagation is how the model makes a prediction.
Backpropagation is how the model learns from its mistakes.

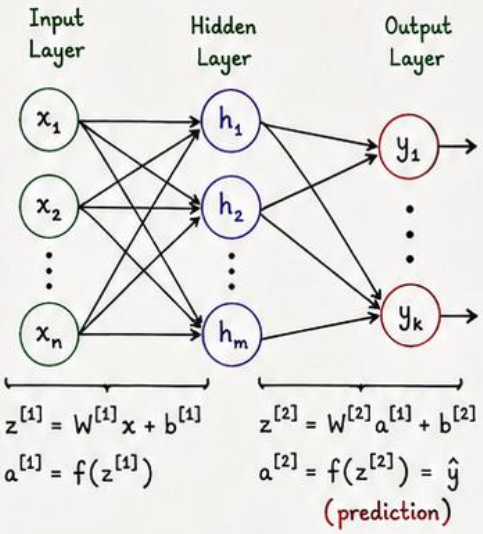
★ 4.1 FORWARD PROPAGATION

Data flows from input layer → hidden layer(s) → output layer to produce the final output.

For a single neuron:

$$z = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b$$
$$a = f(z) \quad (a = \text{activation/output})$$

z = weighted sum (linear part)
a = activation (after applying activation function)
f( ) = activation function



Notations

x	=	input vector
W	=	weights
b	=	bias vector
z	=	linear output
a	=	activation
$\hat{y}$	=	predicted output
y	=	actual output

★ 4.2 BACKPROPAGATION (OVERVIEW)

Backpropagation works backward from output to input layer to update weights and biases to minimize error.

Backpropagation Steps

1. Compute loss using predicted and actual output.
2. Calculate error (gradient) at output layer.
3. Propagate error backward through hidden layers.
4. Update weights and biases using gradients.

Goal: Minimize loss function by adjusting weights and biases.

Why it works?
Calculus (Chain Rule) helps to compute how each weight contributes to the final error. ☆

★ 4.3 BACKPROPAGATION (DETAILS)

1. Output Layer Error

- For regression (MSE):

$$\delta^{[L]} = (\hat{y} - y) \odot f'(z^{[L]})$$

- For classification (Cross Entropy with Softmax):

$$\delta^{[L]} = \hat{y} - y$$

⊙ = element-wise multiplication

2. Hidden Layer Error

$$\delta^{[l]} = (W^{[l+1]})^T \delta^{[l+1]} \odot f'(z^{[l]})$$

Where,
l = current layer
L = last layer

3. Gradient (for updates)

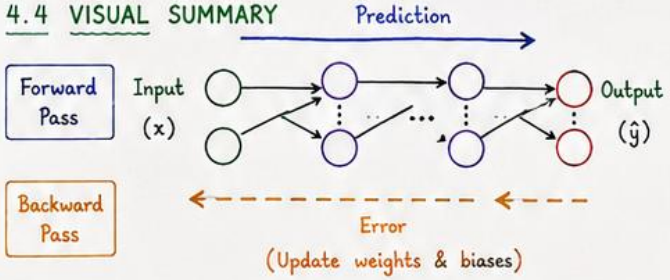
$$dW^{[l]} = \delta^{[l]} (a^{[l-1]})^T$$
$$db^{[l]} = \delta^{[l]}$$

W, b are updated using Gradient Descent

$$W^{[l]} = W^{[l]} - \eta\, dW^{[l]}$$
$$b^{[l]} = b^{[l]} - \eta\, db^{[l]}$$

η = learning rate

4.4 VISUAL SUMMARY



TIPS

- ☆ Understand the flow: Forward → Loss → Backward → Update
- ☆ Activation functions introduce non-linearity.
- ☆ Backpropagation is just an application of the chain rule.
- ☆ Small changes in weights can greatly reduce the loss.
- ☆ Practice with small examples to build strong intuition!

Remember: Forward Propagation makes the prediction, Backpropagation makes the model better. ♥

5. LOSS FUNCTIONS

Loss function measures how wrong the prediction is.

Our goal is to minimize the loss.

★ Lower loss ⟶ better predictions ⟶ better model.

★ 5.1 MEAN SQUARED ERROR (MSE)

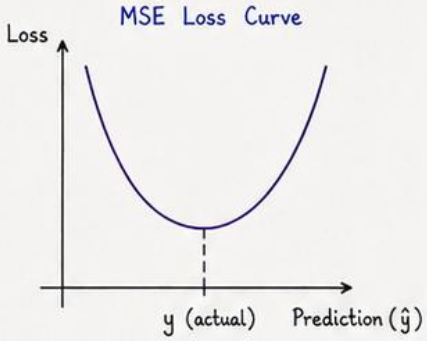
Used for Regression problems.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

y_i = actual value

$\hat{y}_i$ = predicted value

n = number of samples



Pros
- Penalizes large errors more.
- Smooth and differentiable.

Cons
- Sensitive to outliers.
- Not ideal for classification.

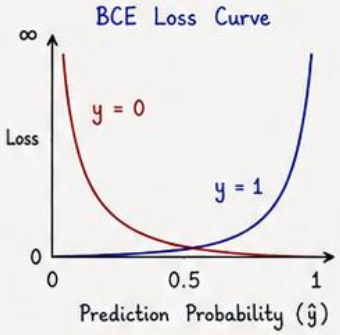
★ 5.2 BINARY CROSS ENTROPY (LOG LOSS)

Used for Binary Classification (0 or 1).

$$BCE = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$$

$y_i \in \{0,1\}$ (actual label)

$\hat{y}_i \in (0,1)$ (predicted probability)



Pros
- Best for binary classification.
- Handles probabilities well.

Cons
- Not used for regression.
- Only for binary output.

★ 5.3 CATEGORICAL CROSS ENTROPY

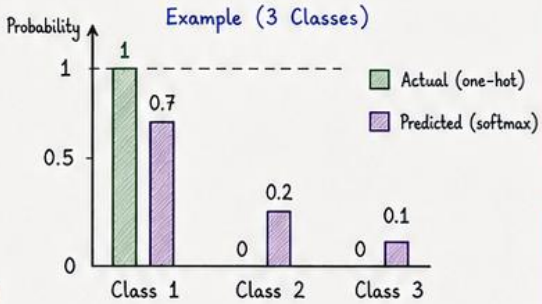
Used for Multi-class Classification (2 or more classes).

$$CCE = -\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{C} y_{ij}\log(\hat{y}_{ij})$$

C = number of classes

y_{ij} = 1 if sample i belongs to class j else 0

$\hat{y}_{ij}$ = predicted probability of class j for sample i



Pros
- Perfect for multi-class classification.
- Works with softmax output.

Cons
- Not for binary (use BCE).
- Sensitive to noisy labels.

QUICK COMPARISON

Loss Function	Used For	Output Type	Range	Best When
MSE	Regression	Continuous	$[0, \infty)$	Predicting numbers (price, height, etc.)
Binary Cross Entropy	Binary Classification	Probability (0 or 1)	$[0, \infty)$	Yes / No, Spam / Not Spam
Categorical Cross Entropy	Multi-class Classification	Probability (multi-class)	$[0, \infty)$	Cat / Dog / Bird classification

Rule of Thumb: Regression → MSE | Binary Classification → BCE | Multi-class Classification → CCE

6. GRADIENT DESCENT & OPTIMIZATION

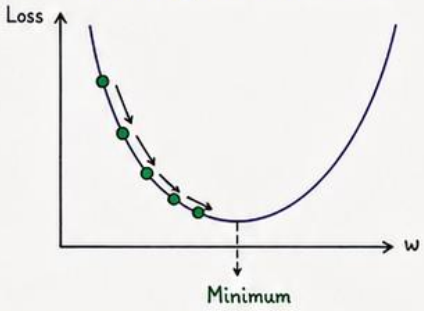
Gradient Descent is an optimization algorithm used to minimize the loss function by updating parameters in the direction of the negative gradient.

★ Goal: Find the set of parameters (weights, biases) that minimize the loss.

★ 6.1 WHAT IS GRADIENT DESCENT ?

- We start with random weights.
- Calculate the gradient (slope) of the loss function.
- Update parameters in the opposite direction of the gradient.
- Repeat until the loss is minimized.

Minimizing the Loss Function



Update Rule

$$w_{t+1} = w_t - \eta \nabla L(w_t)$$

w_t = current weights
η = learning rate
$\nabla L(w_t)$ = gradient of loss function w.r.t weights

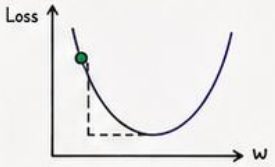
★ 6.2 TYPES OF GRADIENT DESCENT

1. Batch Gradient Descent

Uses the entire dataset to compute gradient.

Update happens once per epoch.

- Stable but slower.

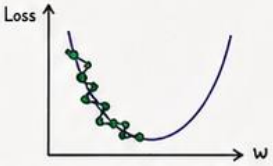


2. Stochastic Gradient Descent

Uses one training example at a time.

Update happens for every example.

- Faster but noisy.

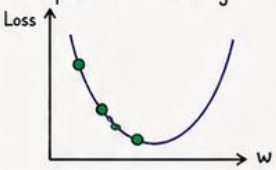


3. Mini-Batch Gradient Descent

Uses a small batch of examples.

Update happens for every mini-batch.

- Good balance between speed and stability.

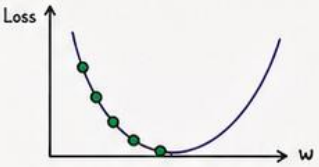


4. Online Gradient Descent

Similar to SGD, used in online/streaming data.

Updates model continuously as data arrives.

- Adapts quickly.



★ 6.3 OPTIMIZATION ALGORITHMS

1. SGD (Vanilla GD)

Simplest form. Good baseline.

$$w_{t+1} = w_t - \eta \nabla L(w_t)$$

2. Momentum

Accelerates in relevant direction & reduces oscillations.

$$v_t = \gamma v_{t-1} + \eta \nabla L(w_t)$$

$$w_{t+1} = w_t - v_t$$

3. RMSProp

Adapts learning rate for each parameter. Good for RNNs.

$$s_t = \beta s_{t-1} + (1-\beta)(\nabla L(w_t))^2$$

$$w_{t+1} = w_t - \frac{\eta}{\sqrt{s_t + \epsilon}} \nabla L(w_t)$$

4. Adam

Combines Momentum and RMSProp. Most popular optimizer.

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)\nabla L(w_t)$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)(\nabla L(w_t))^2$$

$$w_{t+1} = w_t - \eta \frac{\hat{m}t}{\sqrt{\hat{v}_t + \epsilon}}$$

(with bias correction)

TIPS

- Start with Adam (default choice).
- Tune learning rate (η) carefully.
- Too high → overshoot minima.
- Too low → slow convergence.
- Use learning rate schedulers for better results.

★ 6.4 OPTIMIZERS COMPARISON

Optimizer	Memory	Speed	Convergence	Best Used In	Notes
SGD	Low	Fast (per update)	Slow	Large datasets	Simple, may oscillate
Momentum	Medium	Faster	Better	CNNs, Deep Nets	Reduces oscillations
RMSProp	Medium	Fast	Good	RNNs, Non-stationary problems	Handles changing gradients
Adam	High	Fast	Excellent	Most Deep Learning problems	Best default choice

★ Remember: Optimization is the key to training a model efficiently and effectively. ♥

7. OVERFITTING & REGULARIZATION

Overfitting happens when a model learns the training data too well, including the noise, and performs poorly on unseen data.

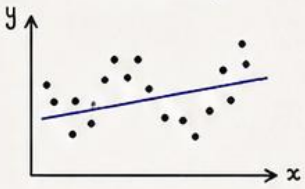
★ Goal: Build a model that generalizes well to new data.

Why it happens?

- Model too complex
- Too many parameters
- Too little training data
- Training for too long

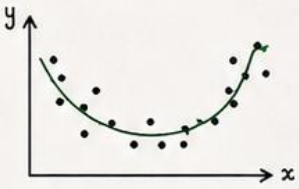
★ 7.1 UNDERSTANDING OVERFITTING

1. Underfitting



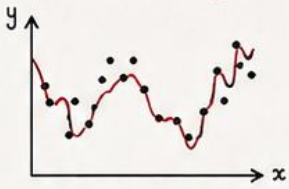
- Model too simple
- High bias, high error

2. Good Fit



- Right complexity
- Low bias, low variance

3. Overfitting



- Model too complex
- Low bias, high variance

Bias vs Variance Trade-off

Increasing complexity

↓

Bias decreases

Variance increases

↓

We need the right balance!

★ 7.2 REGULARIZATION TECHNIQUES

1. L2 Regularization (Ridge)

Adds a penalty equal to the square of the magnitude of weights.

$$L = L_0 + \lambda \sum_{i=1}^{n} w_i^2$$

- Shrinks large weights
- Helps reduce overfitting
- λ = regularization strength

2. L1 Regularization (Lasso)

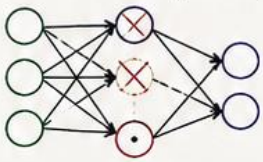
Adds a penalty equal to the absolute value of weights.

$$L = L_0 + \lambda \sum_{i=1}^{n} |w_i|$$

- Can make some weights zero
- Leads to sparse models
- Good for feature selection

3. Dropout

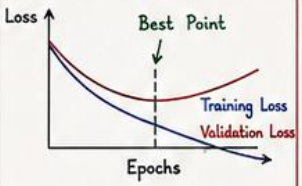
Randomly drops (sets to 0) some neurons during training.



- Prevents co-adaptation
- Acts like training many subnetworks

4. Early Stopping

Stop training when the performance on validation set stops improving.



- Simple but powerful
- Prevents overfitting

5. Batch Normalization

Normalizes the output of a layer to have mean 0 and variance 1.

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

- Stabilizes and speeds up training
- Acts as a regularizer
- Reduces internal covariate shift

6. Data Augmentation

Artificially increases training data by applying transformations.

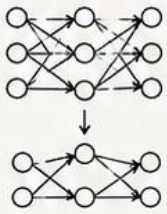


Original Flip Rotate Add Noise

- Improves generalization
- Common in images (flip, crop, rotate, zoom, noise, etc.)

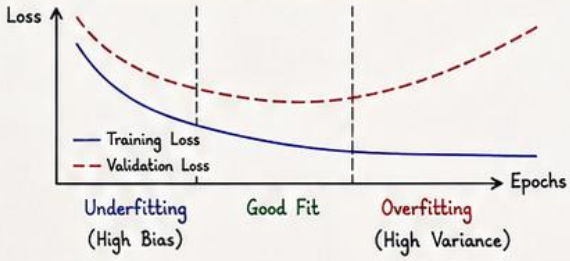
7. Simplify the Model

Use simpler models or reduce the number of parameters.



- Fewer layers
- Fewer neurons
- Remove unnecessary features

★ 7.3 TRAINING CURVE (DETECTION)



Observation	Training Loss	Validation Loss	Conclusion
Both high	High	High	Underfitting (Increase model capacity)
Both low	Low	Low	Good Fit (Model generalizes well)
Train low, Val high	Low	High	Overfitting (Apply Regularization)



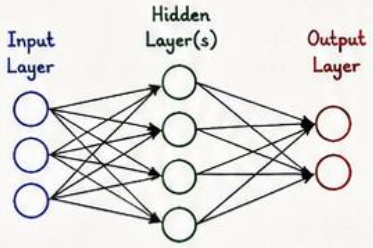
Key Takeaway: Don't just fit the data, understand it. Regularization helps your model generalize and perform well on unseen data.

8. NEURAL NETWORK ARCHITECTURES

Different architectures are designed to handle different types of data and learning tasks effectively.

1. FEEDFORWARD NEURAL NETWORK (FNN)

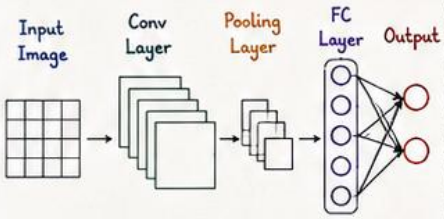
- Data flows only forward, from input layer to output layer.
- No cycles or loops.



Used for: Classification, Regression, Tabular data

2. CONVOLUTIONAL NEURAL NETWORK (CNN)

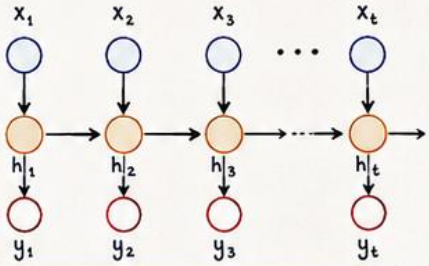
- Uses convolutional layers to extract spatial features.
- Best for image and video data.



Used for: Image recognition, Object detection, Medical imaging

3. RECURRENT NEURAL NETWORK (RNN)

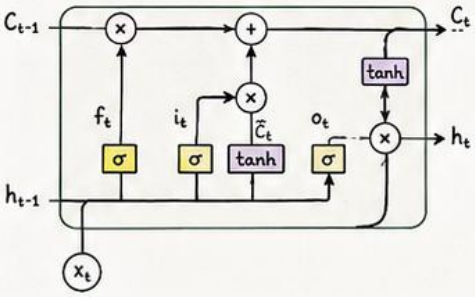
- Designed for sequential data.
- Has loops to maintain hidden state (memory).



Used for: Text, Speech, Time series, Language modeling

4. LSTM (LONG SHORT-TERM MEMORY)

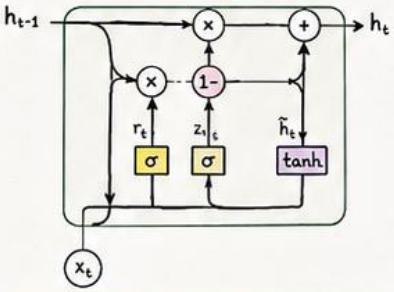
- Overcomes vanishing gradient problem of RNN.
- Uses gates to control information flow.



Used for: Long sequences, Translation, Video analysis

5. GRU (GATED RECURRENT UNIT)

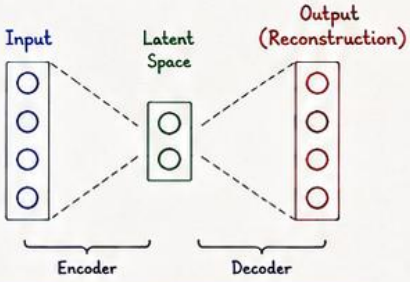
- Simplified version of LSTM.
- Has reset and update gates.



Used for: Similar to LSTM but faster and less complex

6. AUTOENCODER

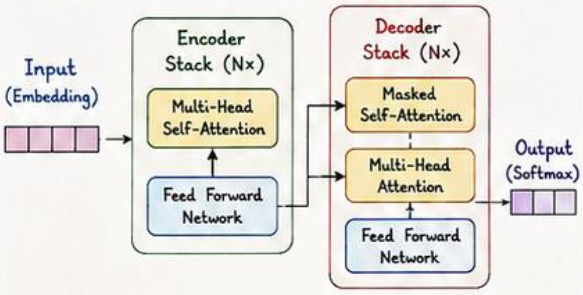
- Unsupervised learning model.
- Learns to compress data and reconstruct it back.



Used for: Dimensionality reduction, Denoising, Anomaly detection

7. TRANSFORMER (ATTENTION BASED MODEL)

- Uses self-attention mechanism instead of recurrence.
- Highly parallelizable, works great on long sequences.



Used for: NLP (BERT, GPT), Translation, Summarization

ARCHITECTURE COMPARISON

Architecture	Best For	Handles Sequence?	Captures Spatial Info?	Training Speed
FNN	Tabular Data	No	No	Fast
CNN	Images / Videos	No	Yes	Medium
RNN	Sequences	Yes	No	Slow
LSTM	Long Sequences	Yes	No	Slow
GRU	Sequences	Yes	No	Faster
Autoencoder	Unsupervised Learning	No	No	Medium
Transformer	NLP / Long Sequences	Yes	No	Fast (Parallel)

Key Takeaway: The choice of architecture depends on the type of data and the problem you are solving.
Understand the data → Choose the right architecture → Train effectively → Get better results. ☆

9. CONVOLUTIONAL NEURAL NETWORKS (CNN)

CNNs are designed to automatically learn spatial hierarchies of features from images using convolution operations.

Why CNNs?

- Exploit spatial structure of data
- Fewer parameters
- Translation invariance
- Great for images & videos

★ 9.1 CONVOLUTION OPERATION

- A small filter (kernel) slides over the input and computes element-wise multiplication and sum.

Output size formula:

$$O = \frac{(N - F + 2P)}{S} + 1$$

Where,
N = input size, F = filter size
P = padding, S = stride

Input Image (5x5)

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

*

Filter / Kernel (3x3)

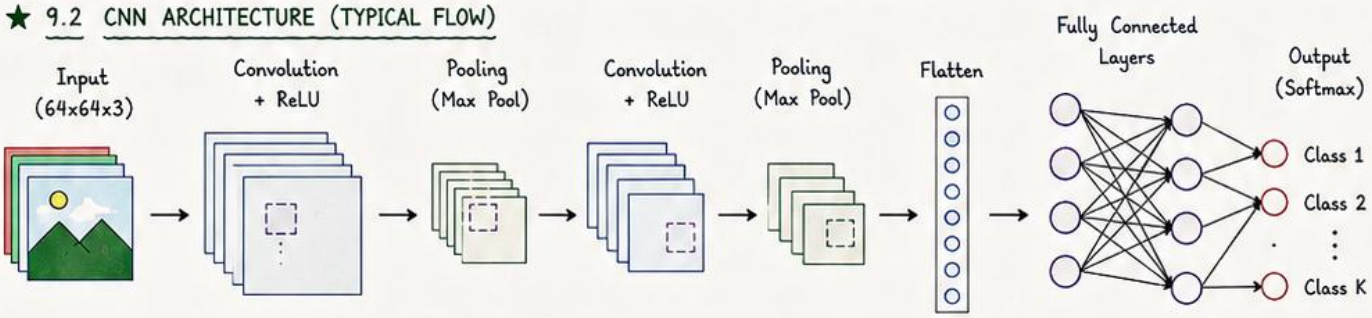
1	0	-1
1	0	-1
1	0	-1

=

Feature Map (Output 3x3)

0	-1	-1
3	0	-3
3	1	-1

★ 9.2 CNN ARCHITECTURE (TYPICAL FLOW)



Conv: Extract features → Pool: Reduce spatial size → FC: Learn high level patterns → Softmax: Predict classes

★ 9.3 COMMON LAYERS IN CNN

1. Convolution Layer: Applies filters to extract features.
2. Activation Layer (ReLU): Introduces non-linearity.
3. Pooling Layer (Max/Average): Reduces spatial size.
4. Batch Normalization: Normalizes activations, speeds up training.
5. Dropout: Prevents co-adaptation.
6. Fully Connected Layer: Performs final classification.
7. Softmax Layer: Converts scores to probabilities.

★ 9.4 POOLING EXAMPLE (MAX POOLING 2x2, STRIDE=2)

Input Feature Map (4x4)

1	3	2	0
4	6	6	1
0	2	1	2
3	1	0	5

Max Pool 2x2, s=2 →

Output (2x2)

6	2
3	5

★ 9.5 POPULAR CNN ARCHITECTURES

Architecture	Year	Key Idea	Best For
LeNet-5	1998	First CNN for digit recognition	Handwritten digits
AlexNet	2012	Deep CNN, ReLU, Dropout	ImageNet classification
VGGNet	2014	Very deep with 3x3 filters	High accuracy tasks
GoogLeNet (Inception)	2014	Inception modules	Efficient feature extraction
ResNet	2015	Residual (skip) connections	Very deep networks
EfficientNet	2019	Scales depth, width, resolution	High accuracy with less compute

KEY TAKEAWAYS

- ☆ CNNs exploit spatial structure of images.
- ☆ Convolution extracts local patterns.
- ☆ Pooling reduces size and computation.
- ☆ Deeper networks learn more complex features.
- ☆ Use ReLU, BatchNorm, Dropout for better training.
- ☆ Softmax gives final class probabilities.

Remember: CNNs transform raw pixels into meaningful features and make the model focus on "WHAT" is important, not "WHERE" it is.

10. RECURRENT NEURAL NETWORKS (RNN)

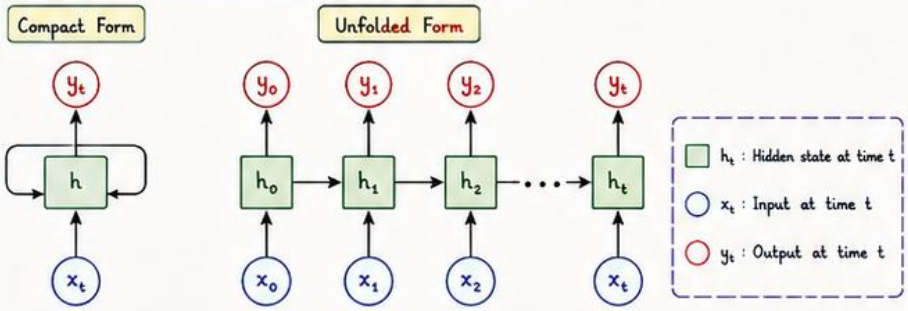
RNNs are designed to handle sequential data by maintaining a hidden state (memory) that captures information from previous time steps.

Why RNNs?

- Handle variable length sequences
- Capture temporal dependencies
- Share parameters across time
- Ideal for text, speech & time series

Goal: Process sequences of any length by remembering what came before.

★ 10.1 RECURRENT NEURAL NETWORK UNFOLDED



10.2 RNN EQUATIONS

Core Equations

$$h_t = f(W_{hx} x_t + W_{hh} h_{t-1} + b_h)$$

$$y_t = g(W_{yh} h_t + b_y)$$

h_t = hidden state at time t
x_t = input at time t
y_t = output at time t
W_{hx}, W_{hh}, W_{yh} = weight matrices
b_h, b_y = bias vectors
f, g = activation functions (tanh, ReLU, softmax)

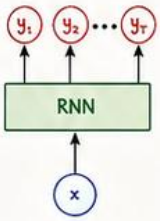
10.3 TYPES OF RNN OUTPUTS

1. One-to-One (Image Classification)



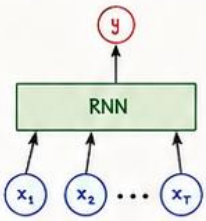
Single input
Single output

2. One-to-Many (Image Captioning)



Single input
Multiple outputs

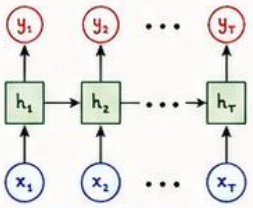
3. Many-to-One (Sentiment Analysis)



Multiple inputs
Single output

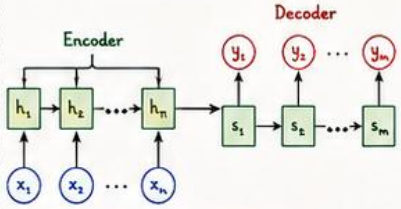
10.4 MANY-TO-MANY OUTPUTS

a) Many-to-Many (Synchronized) (POS Tagging)



Input and output sequences are of the same length

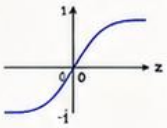
b) Many-to-Many (Encoder-Decoder) (Machine Translation)



Input and output sequences can have different length

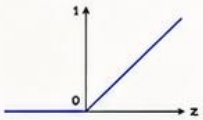
10.5 ACTIVATION FUNCTIONS IN RNN

tanh
Range: (-1, 1)



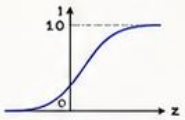
Most commonly used in RNNs

ReLU
Range: [0, ∞)



Faster training but can be less stable

sigmoid
Range: (0, 1)

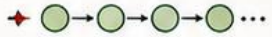


Useful for gates and outputs

10.6 VANISHING & EXPLODING GRADIENTS

Vanishing Gradients

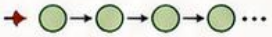
Gradients become too small as they flow back.



Tiny gradients
(Model forgets long-term info)

Exploding Gradients

Gradients grow too large as they flow back.



Huge gradients
(Model becomes unstable)

Solutions: Use LSTM / GRU, Gradient Clipping, Proper Initialization, Layer Normalization

10.7 RNN VARIANTS COMPARISON

Model	Key Idea	Handles Long-term Dependencies?	Training Speed	Complexity	Best Used In
RNN	Basic recurrent connections	No (Vanishing problem)	Slow	Low	Simple sequences
LSTM	Uses gates (input, forget, output)	Yes	Medium	High	Long sequences, NLP, Time series
GRU	Simplified LSTM (update, reset gates)	Yes	Faster	Medium	Real-time apps, Chatbots
BiRNN	Processes sequence in both directions	Yes	Slower	Medium	Speech, NER, Text understanding

SUMMARY

- ☑ RNNs handle sequential data using hidden states.
- ☑ They suffer from vanishing and exploding gradients.
- ☑ LSTM and GRU solve long-term dependency problem.
- ☑ Choose architecture based on data and task.
- ☑ Better architecture → Better performance.

Key Takeaway: RNNs remember the past to predict the future. Understanding sequences is the key to solving real-world problems. ☆

11. TRANSFORMER (ATTENTION IS ALL YOU NEED)

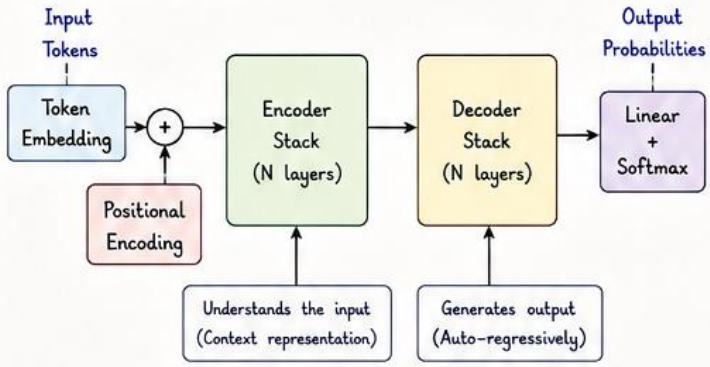
Transformers use self-attention mechanism to understand relationships between all words in a sequence simultaneously.

Why Transformers?

- Captures global context
- Highly parallelizable
- Better for long sequences
- State-of-the-art in NLP & beyond

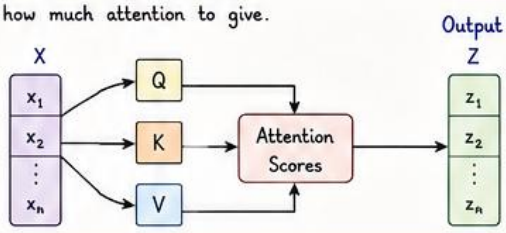
★ **Goal:** Capture long-range dependencies efficiently and enable parallel training.

11.1 THE BIG PICTURE



11.2 SELF-ATTENTION MECHANISM

Each word looks at every other word and decides how much attention to give.

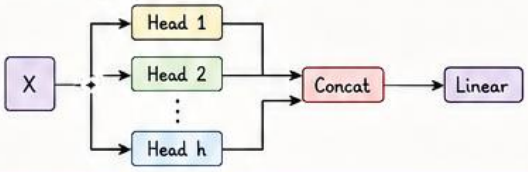


$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Q = Query K = Key V = Value d_k = dimension of keys

11.3 MULTI-HEAD ATTENTION

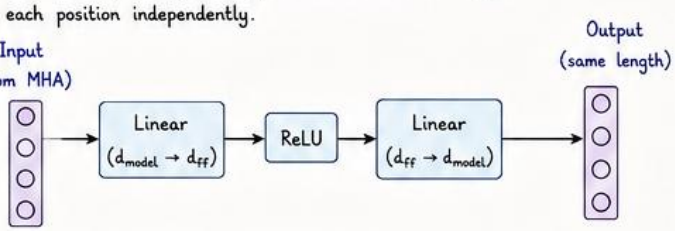
Instead of one attention, use multiple heads to focus on different types of relationships.



$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)\, W^O$$
$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

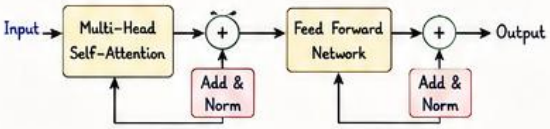
11.4 POSITION-WISE FEED FORWARD NETWORK

After attention, a simple fully connected network is applied to each position independently.



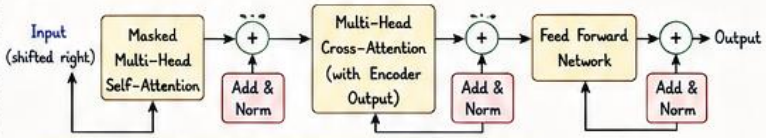
Adds non-linearity and helps in learning complex patterns.

11.5 ENCODER LAYER (REPEATED N TIMES)



Add & Norm = Residual Connection + Layer Normalization

11.6 DECODER LAYER (REPEATED N TIMES)



Masked Self-Attention prevents looking at future tokens.
Cross-Attention helps decoder focus on encoder output.

11.7 POSitional Encoding

Since transformers have no recurrence, we add positional information.

Two common ways:

1. Sinusoidal Encoding (Original)

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$
$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

2. Learned Embedding (Trainable)

A learnable vector added to token embedding.

Example ($d_{model} = 4$)

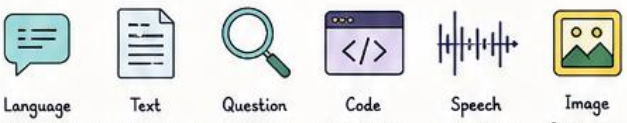
Pos	Positions 0	1	2	3
	PE			
0	0.00	1.00	0.00	1.00
1	0.84	0.54	0.01	1.00
2	0.91	-0.42	0.02	1.00
3	0.14	-0.99	0.03	1.00
...	...	...	...	...

Positional Encoding is added to Token Embedding.

11.8 TRANSFORMER vs RNN

Aspect	RNN	Transformer
Dependency Capture	Sequential (step-by-step)	Global (all positions at once)
Training	Hard to parallelize	Highly parallelizable
Long Sequences	Struggles (vanishing gradients)	Handles well with attention
Speed	Slower	Faster
Memory	High (sequential states)	More efficient
Best Use Cases	Short sequences, streaming	Long sequences, NLP, CV, etc.

11.9 COMMON APPLICATIONS



KEY TAKEAWAYS

- ☆ Transformers use self-attention to capture global context.
- ☆ Multi-head attention allows the model to focus on different parts of the input.
- ☆ Encoder understands, Decoder generates.
- ☆ Positional Encoding gives the model information about order.
- ☆ Transformers are faster, scalable, and more powerful.

Remember: Transformers changed the game! Attention is all you need. ❤

12. TRANSFORMER DECODED

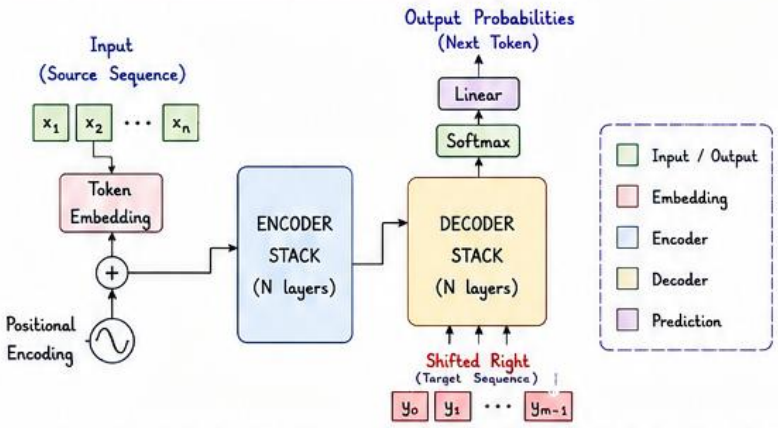
Transformers are built using attention, position awareness and feed forward layers to understand and generate sequences.

Why Transformers?

- Capture long range dependencies
- Parallel processing (faster training)
- More efficient than RNNs
- Backbone of modern NLP models

★ **Goal:** Build a model that understands context and relationships at scale.

12.1 TRANSFORMER ARCHITECTURE (OVERVIEW)



12.2 SCALED DOT-PRODUCT ATTENTION

Given Query (Q), Key (K), Value (V):

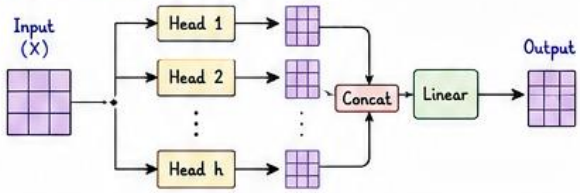
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Q = Query matrix
K = Key matrix
V = Value matrix
d_k = dimension of keys

Why scale by $\sqrt{d_k}$?

To prevent large dot products that push softmax into regions with extremely small gradients.

12.3 MULTI-HEAD ATTENTION



$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)\, W^O$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$

12.4 POSITIONAL ENCODING

Since transformers have no recurrence, positional encoding adds information about token position.

Two common types:

1. Sinusoidal (Original)

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

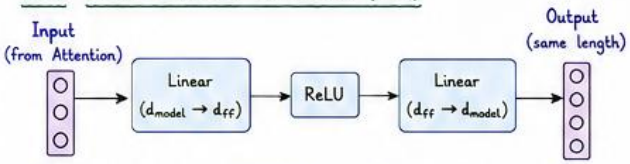
$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

Pos	PE_1	PE_2	...	PE_d
0	0.00	1.00	...	0.02
1	0.84	0.54	...	0.99
2	0.91	-0.42	...	-0.07
3	0.14	-0.99	...	-0.03
...	...	...	...	...

2. Learned (Trainable)

A learnable vector is added to token embedding.

12.5 FEED FORWARD NETWORK (FFN)

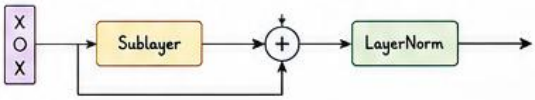


Applied position-wise and identically to all positions.

Typical sizes: d_{model} = 512/768/1024

d_{ff} = 2048/3072/4096 ($\approx 4 \times d_{model}$)

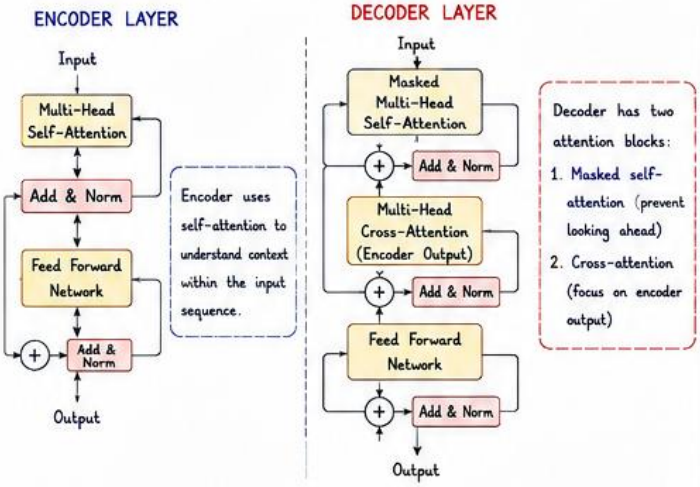
12.6 LAYER NORMALIZATION & RESIDUAL CONNECTION



$$\text{Output} = \text{LayerNorm}(X + \text{Sublayer}(X))$$

- Residual connection helps gradients flow.
- LayerNorm stabilizes training and speeds up convergence.

12.7 ENCODER vs DECODER LAYER



12.8 MODEL COMPLEXITY (PER LAYER)

Component	Time Complexity	Space Complexity (Memory)
Self-Attention	$O(n^2 d)$	$O(n^2)$
Cross-Attention	$O(n\, m\, d)$	$O(n\, m)$
Feed Forward	$O(n\, d\, d_{ff})$	$O(n\, d_{ff})$

n = sequence length, d = model dim, d_{ff} = feed forward dim, m = encoder output length

12.9 COMMON APPLICATIONS



KEY TAKEAWAYS

- ☆ Transformers rely on attention, not recurrence.
- ☆ Self-attention captures global context.
- ☆ Positional encoding gives order information.
- ☆ Residual connections + LayerNorm = stable training.
- ☆ Transformers are the foundation of modern AI models.

Remember: Attention helps you focus on what matters. Transformers help you see the bigger picture. ♥

13. TRANSFORMER IN DEPTH

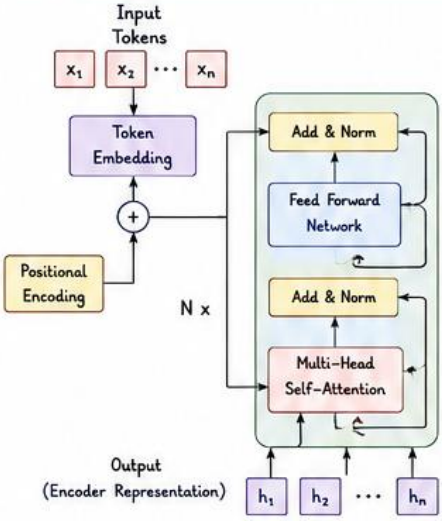
Transformers are the backbone of modern NLP. Let's go deeper!

Goal: Deeply understand how transformers work, why they work so well and where they are used.

Why Transformers Dominate?

- Capture long-range dependencies
- Parallelizable (faster training)
- Scale to huge models & datasets
- State-of-the-art in NLP, CV, Speech & more

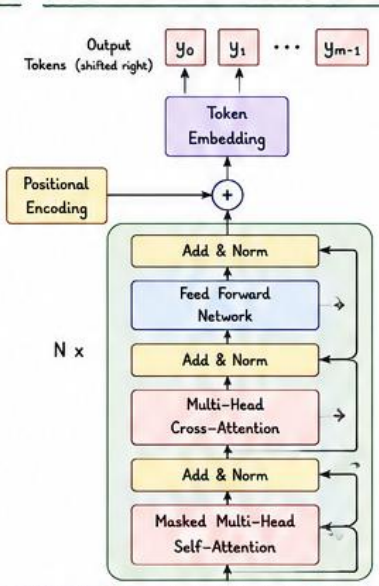
13.1 THE TRANSFORMER BLOCK (ENCODER)



Components

1. Multi-Head Self-Attention
2. Add & Norm (Residual + LayerNorm)
3. Feed Forward Network (FFN)
4. Add & Norm

13.2 THE TRANSFORMER BLOCK (DECODER)



Components

1. Masked Multi-Head Self-Attention
2. Add & Norm
3. Cross-Attention
4. Add & Norm
5. Feed Forward Network
6. Add & Norm

13.3 SCALED DOT-PRODUCT ATTENTION

Inputs: Q (query), K (key), V (value)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Why scale by $\sqrt{d_k}$?

- Prevents softmax from having extremely small gradients.
- Helps in stable and efficient training.

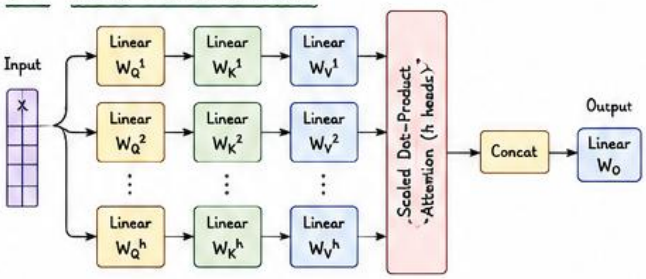
Example

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad V = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$d_k = 2$

$$\text{Attention} = \text{softmax}\left(\frac{QK^T}{\sqrt{2}}\right)V = \text{softmax}\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix} V$$

13.4 MULTI-HEAD ATTENTION



Multi-head attention allows the model to attend to information from different representation subspaces at different positions.

13.5 POSITIONAL ENCODING

Since transformers have no recurrence, we add positional information.

Two types:

1. Sinusoidal (Fixed)
2. Learned (Trainable)

Sinusoidal Encoding (Example →

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

Added to token embeddings.

Example ($d_{model} = 4$)

pos	PE_0	PE_1	PE_2	PE_3
0	0.00	1.00	0.00	1.00
1	0.84	0.54	0.01	1.00
2	0.91	-0.42	0.02	1.00
3	0.14	-0.99	0.03	1.00
...	...	...	...	...

13.6 MASKED MULTI-HEAD SELF-ATTENTION

In the decoder, we mask future tokens to prevent cheating!

Look-ahead Mask (Example)

	0	1	2	3
0	1	0	0	0
1	1	1	0	0
2	1	1	1	0
3	1	1	1	1

Allowed (can attend)

Masked (cannot attend)

1 = Visible, 0 = Masked

13.7 BENEFITS OF TRANSFORMERS

- **Parallelizable** — Train much faster than RNNs.
- **Long-range Dependencies** — Attention can connect distant words.
- **Scalability** — Works well with large datasets & models.
- **Versatility** — Used in NLP, CV, Speech, RL & more.

13.8 REAL-WORLD USE CASES

ChatGPT (Conversational AI)	Google Translate (Language Translation)	BERT (Search & QA)
GitHub Copilot (Code Generation)	Whisper (Speech Recognition)	DALL-E (Text to Image)

13.9 SUMMARY

- [x] Transformers use attention, not recurrence.
- [x] Self-attention captures relationships within the sequence.
- [x] Positional Encoding adds order information.
- [x] Multi-head attention focuses on different parts.
- [x] Feed Forward Networks add non-linearity.
- [x] Transformers are scalable, efficient and state-of-the-art.

Key Takeaway: Transformers changed the game by replacing recurrence with attention. They look at the whole picture, learn faster and understand better!

14. TRAINING TRANSFORMERS

Training a transformer means teaching it to predict the **next token** by minimizing the **error** between predicted and actual tokens.

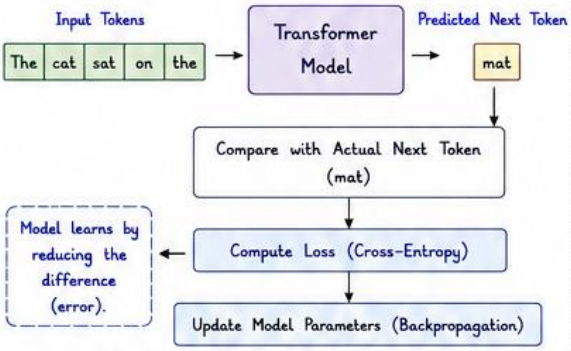
> ★ **Goal:** Minimize prediction error so model learns patterns, understands context and generalizes to new data.

Why Training Matters?

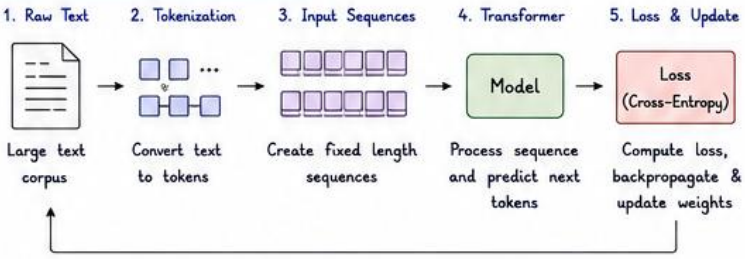
- Learns patterns from massive text data
- Improves understanding of context & semantics
- Enables accurate next-token prediction
- Makes the model generalize to unseen data
- Backbone for all downstream tasks

14.1 TRAINING OBJECTIVE

Transformers are trained using Next Token Prediction.



14.2 THE TRAINING PIPELINE



Training Loop (Repeat Many Times)

Forward Pass → Compute Loss → Backpropagation → Update Weights

14.3 LOSS FUNCTION (CROSS-ENTROPY)

Cross-entropy measures the difference between predicted probability distribution and actual next token.

For a single position:

$$\text{Loss} = -\sum_{i=1}^{V} y_i \log(\hat{y}_i)$$

Where,

- V = vocabulary size
- y_i = actual probability (one-hot)
- $\hat{y}_i$ = predicted probability

Intuition

Lower loss → model is confident and correct.

Higher loss → model is wrong or unsure.

14.4 OPTIMIZATION

We use optimizers to update model parameters and minimize loss.

Popular Optimizers:

Optimizer	Key Idea	Used By
Adam	Adaptive learning rate	Most common
AdamW	Adam + Weight Decay	GPT, BERT, T5
SGD + Momentum	Momentum to accelerate	Older models
Adafactor	Memory efficient	T5, Switch Transformer

Key Hyperparameters

- Learning Rate (e.g. 1e-4)
- Batch Size (e.g. 32, 64)
- Optimizer (AdamW)
- Weight Decay (e.g. 0.01)
- Warmup Steps
- Total Training Steps

14.5 REGULARIZATION TECHNIQUES

Dropout
Randomly drops neurons during training.

Weight Decay
Penalizes large weights to prevent overfitting.

Label Smoothing
Prevents model from being over-confident.

Early Stopping
Stops training when validation loss stops improving.

14.6 DATA & SCALING

Data Requirements

- Huge amounts of text data (GBs to TBs).
- Diverse domains improve generalization.
- High quality data > more important.

Scaling Factors

- More Data → Better Learning
- Larger Model → More Capacity
- More Compute → Better Performance

Scaling Laws (Simplified)

More Data → Lower loss

Bigger Model → Lower loss

More Compute → Lower loss

But returns diminish! Find the right balance.

14.7 TRAINING AT SCALE

Distributed Training
Use multiple GPUs/TPUs across machines.

Data Parallelism
Same model on different data shards.

Model Parallelism
Split model across multiple devices.

Mixed Precision Training
Use FP16/BF16 for faster training & less memory.

14.8 EVALUATION DURING TRAINING

Metrics

- ☑ Training Loss
- ☑ Validation Loss
- ☑ Perplexity (PPL)
- ☑ Accuracy (for tasks)

Typical Curves



14.9 COMMON CHALLENGES IN TRAINING

Challenge	Why it Happens	What Helps
Overfitting	Model memorizes training data	More data, regularization, early stopping
Underfitting	Model too small or not trained enough	Bigger model, more training
Vanishing/Exploding Gradients	Improper initialization or deep model	LayerNorm, Residual Connections, good init
Slow Training	Large model, limited compute	Better hardware, optimization, mixed precision
High Memory Usage	Large batch size, long sequences	Gradient checkpointing, smaller batch, efficient attention

14.10 SUMMARY

- ☑ Transformers are trained using next token prediction.
- ☑ Cross-entropy loss is minimized using backpropagation.
- ☑ Optimizers update weights to reduce loss.
- ☑ Regularization prevents overfitting and improves generalization.
- ☑ Scaling data, model size and compute improves performance.
- ☑ Efficient training at scale enables powerful real-world models.

REMEMBER

Training = Data + Model + Compute + Time

Better data + smart training = Smarter models! ♥

15. REAL-WORLD APPLICATIONS OF TRANSFORMERS

Transformers are used everywhere! Let's see how they power modern AI systems.

Goal: Understand where transformers are used and why they are the best choice for these problems.

Why Transformers?

- Handle long-range dependencies
- Understand context and meaning
- Work well with large-scale data
- Adapt to many tasks and domains
- Continually improving with research

15.1 NATURAL LANGUAGE PROCESSING (NLP)

Application	Example Models	What Transformers Do
Machine Translation	Google Translate, DeepL, mBERT	Translate text from one language to another.
Text Summarization	BART, T5, PEGASUS	Summarize long articles or documents.
Question Answering	BERT, RoBERTa, ALBERT	Find precise answers from large passages.
Sentiment Analysis	BERT, DistilBERT, XLNet	Understand positive, negative or neutral sentiment.
Text Generation	GPT, LLaMA, PaLM	Generate human-like and context-aware text.

15.2 COMPUTER VISION (CV)

Application	Example Models	What Transformers Do
Image Classification	ViT, DeiT, Swin Transformer	Classify images with high accuracy.
Object Detection	DETR, Deformable DETR	Detect and locate objects in images.
Image Segmentation	SegFormer, Swin-UNet	Pixel-level understanding of images.
Video Understanding	Video Transformer, TimeSformer	Understand actions and events in videos.
Medical Imaging	Swin Transformer, TransUNet	Analyze X-ray, MRI, CT scans for diagnosis.

15.3 SPEECH & AUDIO

Application	Example Models	What Transformers Do
Speech Recognition	Whisper, Wav2Vec 2.0, Conformer	Convert speech to accurate text.
Text-to-Speech	FastSpeech 2, VITS	Generate natural sounding speech from text.
Speech Translation	SeamlessM4T, SpeechT5	Translate speech across different languages.
Speaker Recognition	ECAPA-TDNN + Transformers	Identify and verify speakers.

15.4 RECOMMENDATION SYSTEMS

Application	Example Models	What Transformers Do
Product Recommendation	SASRec, BERT4Rec, CoRel	Recommend products based on user behavior.
Content Recommendation	YouTube DNN, TransNet	Suggest videos, articles or content.
Personalized Search	BERT, TFRS	Rank results based on user intent.
News Feed Ranking	Transformer Ranker, DeepRec	Rank and show most relevant posts.

15.5 MULTI-MODAL APPLICATIONS

Application	Example Models	What Transformers Do
Image + Text Understanding	CLIP, ViLT, BLIP-2	Understand both image and text together.
Visual Question Answering (VQA)	LLaVA, ViLT, BLIP	Answer questions about images.
Document AI	LayoutLM, Donut, DocFormer	Extract info from scanned documents, forms, invoices.
Code + Text Understanding	CodeBERT, PLBART, StarCoder	Understand and generate code with natural language.

15.6 OTHER EMERGING APPLICATIONS

Application	Example Models	What Transformers Do
Scientific Research	SciBERT, Galactica, BioGPT	Analyze papers, generate hypotheses, summarize.
Finance	FinBERT, Longformer, BloombergGPT	Analyze market news, predict trends.
Robotics	RT-1, PaLM-E, Perceiver IO	Understand environment and plan actions.
Healthcare NLP	BioBERT, ClinicalBERT, Med-PaLM	Analyze medical records, predict health outcomes.

15.7 WHY TRANSFORMERS WORK SO WELL?

- ☑ Self-attention captures all relationships in data.
- ☑ Parallelizable → trains faster on modern hardware.
- ☑ Scales to huge models and datasets.
- ☑ Transfer learning makes them adaptable to many tasks.
- ☑ State-of-the-art results across almost every domain.

15.8 CHALLENGES & LIMITATIONS

- ⚠ High computational cost (training is expensive).
- ⚠ Requires large amounts of data.
- ⚠ Memory intensive for long sequences.
- ⚠ Can generate biased or incorrect outputs.
- ⚠ Not always interpretable (black box).

15.9 THE FUTURE

- ☆ More efficient models (smaller, faster, greener).
- ☆ Better long-context understanding.
- ☆ Stronger multi-modal integration.
- ☆ Personalized AI for everyone.
- ☆ Transformers + reasoning = AGI potential!

15.10 SUMMARY

- ☑ Transformers are the backbone of modern AI.
- ☑ Used in NLP, CV, Speech, Recommenders and more.
- ☑ Handle complex patterns and long-range dependencies.
- ☑ Continuously evolving and improving.
- ☑ They are shaping the future of technology!

WHEREVER THERE IS DATA, TRANSFORMERS CAN HELP!

Language · Vision · Speech · Recommenders · Documents · Robotics · Finance · Healthcare

Key Takeaway: Transformers changed how AI understands the world. They learn from data, find meaning, and create impact everywhere!

TRANSFORMER CHEAT SHEET

— THE ULTIMATE QUICK REFERENCE —

Remember: Attention is all you need.

1. WHAT IS A TRANSFORMER?

Transformer is a deep learning model based on self-attention mechanism to understand relationships between all words in a sequence simultaneously.

Goal: Capture long-range dependencies efficiently and enable parallel training.

2. CORE IDEA



- ☑ No recurrence (No RNN)
- ☑ Fully parallelizable
- ☑ Captures global context

3. KEY FORMULA

Scaled Dot-Product Attention

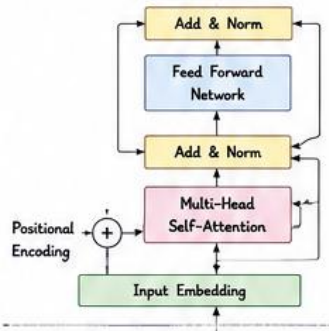
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Q = Query matrix K = Key matrix
V = Value matrix d_k = dimension of keys

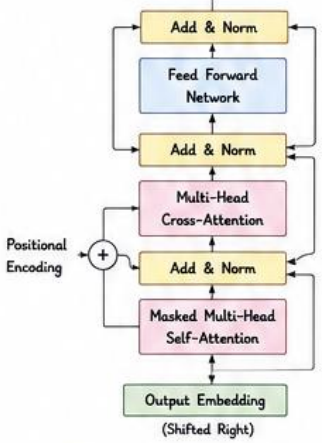
Purpose: Measures how much focus to give from query to different keys.

4. ARCHITECTURE OVERVIEW

ENCODER (N layers)



DECODER (N layers)

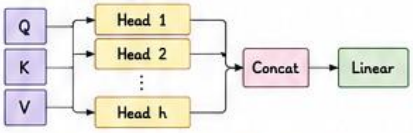


5. ATTENTION VARIANTS

a) Scaled Dot-Product Attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

b) Multi-Head Attention



$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\, W^O$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$

c) Masked Multi-Head Self-Attention (in Decoder)

Look-ahead mask prevents attending to future tokens.

Query Position (i) \ Key Position (j)	0	1	2	3	4
0	1	1	1	0	0
1	1	1	0	0	0
2	1	1	0	0	0
3	1	1	1	0	0
4	1	1	1	1	0

1 = Can attend
0 = Cannot attend

6. POSITIONAL ENCODING

Since transformers have no recurrence, positional encoding adds information about token position.

a) Sinusoidal (Original)

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

b) Learned (Trainable)

A learnable vector is added to token embedding.

Example (d_{model} = 4)

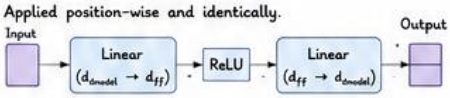
Pos	PE_0	PE_1	PE_2	PE_3
0	0.00	1.00	0.00	1.00
1	0.84	0.54	0.01	1.00
2	0.91	-0.42	0.02	1.00
3	0.14	-0.99	0.03	1.00
...	...	...	...	...

Why Positional Encoding?

- ☑ Adds order information
- ☑ Helps model understand sequence
- ☑ Enables generalization to longer sequences

7. FEED FORWARD NETWORK

Applied position-wise and identically.

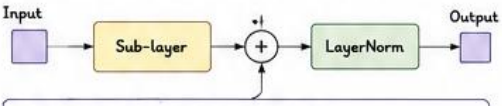


Typical sizes:

d_{model} = 512 / 768 / 1024

d_{ff} = 4 × d_{model} (e.g., 2048 / 3072 / 4096)

8. LAYER NORMALIZATION & RESIDUAL



- ✖ Residual connection helps gradients flow.
- ✖ LayerNorm stabilizes training and speeds up convergence.

9. COMPLEXITY (PER LAYER)

Component	Time Complexity	Space Complexity (Memory)
Self-Attention	$O(n^2 d)$	$O(n^2)$
Cross-Attention	$O(n\, m\, d)$	$O(n\, m)$
Feed Forward	$O(n\, d\, d_{ff})$	$O(n\, d_{ff})$

n = sequence length, d = model dim, d_{ff} = feed forward dim, m = encoder output length

10. TRANSFORMER vs RNN

Aspect	RNN	Transformer
Dependency Capture	Sequential (step-by-step)	Global (all positions at once)
Training	Hard to parallelize	Highly parallelizable
Long Sequences	Struggles (vanishing gradients)	Handles well with attention
Speed	Slower	Faster
Memory	High (sequential states)	More efficient
Best Use Cases	Short sequences, streaming	Long sequences, NLP, CV, etc.

11. COMMON APPLICATIONS

- Language Translation
- Text Summarization
- Question Answering
- Code Generation
- Speech Recognition
- Image Captioning
- Recommendation
- Search & Ranking
- Time Series Forecasting
- Scientific Research

12. TRAINING ESSENTIALS

Loss Function	Cross-Entropy Loss
Optimizer	Adam / AdamW
Learning Rate	1e-4 (warmup + decay)
Dropout	0.1 (common)
Batch Size	32 / 64 (depends on GPU)
Warmup Steps	Usually 4000 – 10000
Label Smoothing	0.1 (helps generalization)
Weight Decay	0.01 (common)

13. CHALLENGES & LIMITATIONS

- ⚠ High computational cost ($O(n^2)$ in attention)
- ⚠ Requires large amount of data
- ⚠ Memory intensive for long sequences
- ⚠ Can generate biased or incorrect outputs
- ⚠ Not always interpretable (black box).

14. WHY TRANSFORMERS ARE POWERFUL

- ☑ Self-attention captures all relationships
- ☑ Works well with large-scale data
- ☑ Parallelizable → trains faster
- ☑ Transfer learning makes them versatile
- ☑ State-of-the-art in many domains

15. FUTURE & TRENDS

- ★ More efficient models (smaller, faster, greener)
- ★ Better long-context models
- ★ Stronger multi-modal models
- ★ Personalized AI systems
- ★ Transformers + Reasoning = AGI potential

16. QUICK REFERENCE FORMULAS

1. Attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

2. Multi-Head

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\, W^O$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

3. Positional Encoding (Sinusoidal)

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

4. Layer Norm

$$\text{LayerNorm}(X) = \gamma\,\frac{X - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta$$

μ = mean, σ = std, ϵ = 1e-5
γ, β = learnable params

Notation

n : sequence length
d_{model} : model dimension
d_{ff} : feed forward dimension
h : number of heads
m : encoder output length

Key Takeaway: Transformers changed the game by replacing recurrence with attention. They understand the whole picture, learn faster and scale better!